

Oswin Franco Cervantes

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Technical Profile

Electrical engineering and physics student focused on analog/mixed-signal ICs, RF/microwave systems, and applied electromagnetics, with experience spanning semiconductor device modeling, schematic design, simulation, layout, fabrication, and laboratory validation.

Education

Massachusetts Institute of Technology, Cambridge, MA

Expected May 2028

B.S. Candidate in Physics and Electrical Engineering with Computing

Sept. 2024–Present

Relevant coursework: **MIT EE:** High-Frequency Integrated Circuits (graduate), Electromagnetics (graduate), Solid-State Circuits, Semiconductor Electronic Circuits, Signal Processing, Digital Systems Laboratory (Verilog); **MIT Physics:** Quantum Physics I, Statistical Mechanics, Classical Mechanics II, Vibrations and Waves; **Professional:** IEEE MTT-S IMS2026 RF Boot Camp; **Coursera:** Microwave Engineering and Antennas, RF and Millimeter-Wave Circuit Design

Georgia Tech, Atlanta, GA — Dual Enrollment, Mathematics/Computer Science; GPA: 4.0/4.0

Aug. 2022–May 2024

Selected Analog IC Design Project

22 nm FinFET Amplifier and Control Chip

Spring 2026

MIT Semiconductor Electronic Circuits

Cambridge, MA

- Designed and simulated a 22 nm FinFET CMOS analog amplifier with activation switches and digital control, applying semiconductor device physics and large- and small-signal models.
- Completed the Cadence schematic and physical layout for a course tapeout; used Calibre DRC/LVS to verify fabrication rules and schematic-layout consistency.
- Verified in simulation that the amplifier exceeded 26 dB gain at 1 MHz and 300 mV peak-to-peak output-swing targets while operating 48% below its 500 μ W power limit.

Relevant Experience

Undergraduate Researcher — Prof. William D. Oliver

Sept. 2025–Present

MIT Research Laboratory of Electronics, Engineering Quantum Systems (EQuS) Group

Cambridge, MA

- Design and test niobium coplanar-waveguide LC filters for hybrid quantum hardware; benchmark 2nd-, 4th-, and 8th-order designs with a 100 MHz cutoff and minimal passband insertion loss, with the 8th-order filter achieving 35.91 dB worst-case, 48.52 dB median, and 60.77 dB peak attenuation across the 3–5 GHz stopband.
- Create RF layouts in KLayout and model eigenmodes and S-parameters in Ansys HFSS/Maxwell; automate simulations with PyAEDT, extract parasitics in Q3D, and develop a machine-learning model to optimize higher-order LC-filter parameters.
- Develop reusable components and fabrication-ready GDS layouts in CDES, a Python/gdstk superconducting-circuit framework, using GitHub version control; fabricate niobium-on-silicon devices through cleanroom photolithography.

Hardware Development Engineer (Systems Engineering) Intern

May 2025–Aug. 2025

Amazon Lab126, Advanced Products

Sunnyvale, CA

- Designed and simulated buck, boost, and buck-boost power-regulation circuits in OrCAD Capture and SPICE; laid out PCBs in Cadence Allegro and evaluated efficiency and maximum power point tracking (MPPT) with oscilloscopes and electronic loads.
- Worked cross-functionally to translate ambiguous product requirements into system architecture, block diagrams, schematics, and PCB layout for a multi-IC solar-charging module compatible with Amazon Kindle devices.
- Integrated and debugged ICs over 1^2 C; assembled and tested 0402, 0603, and 0804 SMD prototypes and debug boards to validate system performance; trained new hires on PCB tools.

Avionics Engineer

Jan. 2026–Present

MIT Rocket Team

Cambridge, MA

- Design RF circuits and PCB interconnects in Altium; document electronic parts and specifications; validate HFSS Driven Terminal and SIwave S-parameter models against vector network analyzer measurements.
- Reduce RF component cost by designing custom 5.8 GHz and 915 MHz microstrip quadrature couplers on FR-4 PCBs to replace expensive stripline parts and unsuitable LTCC alternatives.

Technical Skills

Analog/RF Design: 22 nm FinFET CMOS, analog and mixed-signal ICs, solid-state circuits, semiconductor device modeling, large- and small-signal analysis, power conversion, RF and microwave circuits, S-parameters, transmission lines, impedance matching, filters and resonators, SPICE, PCB design, digital and RTL design

Layout, Fabrication, and Test: Cadence Virtuoso/Allegro/AWR Microwave Office, Ansys HFSS/Maxwell/Q3D/Slwave/PyAEDT, Calibre DRC/LVS, Altium, OrCAD, KLayout; GDS layout, niobium-on-silicon photolithography, cleanroom processing, VNA, oscilloscopes, electronic loads, SMD assembly

Programming/HDL: Python, MATLAB, C, Java, Julia, Bash, Verilog, VHDL, Bluespec, RISC-V assembly; NumPy, pandas, gdstk, Git

Honors and Certification

Honors: QuestBridge Scholarship; Amazon Scholar; Jack Kent Cooke Foundation Semifinalist; Stanford Intrologic Scholar

Certification: TestOut Network Pro Certification